

RS003 Week 8 — English Worksheet

Topic: Variable Speed — Hold Shift to Go Fast · **Course:** Pygame Turret · **Time:** about 45 minutes

This week you make the aiming **feel good**. The barrel speeds up while you hold **Shift**, and when you let go of the arrow keys it slowly coasts to a stop with **friction**. You learn about a **velocity** variable that builds up and slows down.

 Words to know: **velocity**, **acceleration**, **friction**, **max speed**, **Shift key**

1 · Predict

Read each piece of code. Before you run it, write what you think will happen.

```
velocity_x = 0
ACCEL = 0.5
velocity_x += ACCEL
gun_tip_x = gun_tip_x + int(velocity_x)
```

velocity_x keeps growing while you hold a key. What happens to the barrel's movement — does it move at a steady speed or get faster?


```
if not keys[pygame.K_LEFT] and not keys[pygame.K_RIGHT]:
    velocity_x = velocity_x * FRICTION    # FRICTION = 0.85
```

When you let go of both keys, the velocity is multiplied by 0.85 every loop. What does that do over time?


```
max_v = MAX_SPEED_FAST if keys[pygame.K_LSHIFT] else MAX_SPEED
```

What does holding Shift change about the top speed?

2 · Run It

 **Type the example, run it, and feel the coast**

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

gun_tip_x = CENTER_X
velocity_x = 0
ACCEL = 0.5
MAX_SPEED = 8
MAX_SPEED_FAST = 16
FRICTION = 0.85

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    max_v = MAX_SPEED_FAST if (keys[pygame.K_LSHIFT] or keys[pygame.K_RSHIFT]) else MAX_SPEED

    if keys[pygame.K_LEFT]:
        velocity_x -= ACCEL
    if keys[pygame.K_RIGHT]:
        velocity_x += ACCEL

    if velocity_x > max_v:
        velocity_x = max_v
    if velocity_x < -max_v:
        velocity_x = -max_v

    if not keys[pygame.K_LEFT] and not keys[pygame.K_RIGHT]:
        velocity_x = velocity_x * FRICTION

    gun_tip_x = gun_tip_x + int(velocity_x)

    screen.fill((135, 206, 235))
    pygame.draw.rect(screen, (0, 100, 0), (CENTER_X - 30, HEIGHT - 80, 60, 40))
    pygame.draw.line(screen, (80, 80, 80), (CENTER_X, HEIGHT - 80), (gun_tip_x, HEIGHT - 200), 10)
    pygame.display.flip()
```

```
clock.tick(60)
```

```
pygame.quit()
```

Hold RIGHT, then let go. Does the barrel stop instantly or coast? Now hold Shift and aim. What changes?

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — The barrel should coast to a stop when you let go, but instead it keeps sliding forever.

```
if keys[pygame.K_LEFT]:
    velocity_x -= ACCEL
if keys[pygame.K_RIGHT]:
    velocity_x += ACCEL
gun_tip_x += int(velocity_x)
```

Hint: nothing slows the velocity down when no key is held.

Write the fixed code (add the friction step):

Why was it wrong? Why does your fix work?

Bug B — The barrel is supposed to have a top speed, but holding a key for a long time makes it shoot across the screen far too fast.

```
if keys[pygame.K_RIGHT]:
    velocity_x += ACCEL
gun_tip_x += int(velocity_x)
```

Hint: the velocity is never capped at a maximum.

Write the fixed code (add the speed limit):

Why was it wrong? Why does your fix work?

Bug C — The program crashes because Pygame is given a position with a decimal point.

```
gun_tip_x = gun_tip_x + velocity_x
pygame.draw.line(screen, GRAY, (CENTER_X, HEIGHT - 80), (gun_tip_x, HEIGHT - 200),
10)
```

Hint: velocity is a decimal; screen positions must be whole numbers.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Tune the friction

Try `FRICTION = 0.5`, then `0.85`, then `0.99`. Run each one and let go after aiming.

Write how each friction value feels (which one is slippery, which one is grippy):

Make Shift mode look different

Add a visual change while Shift is held — for example, make the barrel a different colour.

Describe what you changed and which key state triggers it:

Hint: you already know `keys[pygame.K_LSHIFT]`. Use it to pick a colour the same way the code picks `max_v`.

5 · Make It 📷

Build a turret with feel

Build a program where:

1. the barrel speeds up while held (acceleration),
2. it coasts to a stop when released (friction),
3. Shift gives a faster top speed,
4. it never crashes from decimal positions.

Send a **video** comparing normal aiming and Shift aiming, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I added a velocity variable that ...

Acceleration makes the barrel ...

Friction makes it ... when I let go.

Holding Shift changes the top speed to ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video showing normal aiming and Shift aiming. Talk through it like you are teaching someone. Try to use these words: **velocity**, **acceleration**, **friction**, **max speed**, **Shift**.

1. Aim slowly, then hold Shift and aim fast.
2. Let go of the keys and show the barrel coasting.
3. Read your friction line out loud and say what 0.85 does.
4. Show the difference between two friction values you tried.

Write what you will say in your video. Plan it here before you record.

Submit ✓

Send this worksheet + your normal/Shift comparison video to teacher on KakaoTalk.